Reward design for RL assisted RAG

MSML635 - Reinforcement Learning · Group 5

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1 Background

Retrieval-augmented generation (RAG) hands a language model some text blocks from an external corpus at inference time, so it can answer questions its parameters do not cover [2]. Retrievers are imperfect. When they return the right passages, models usually answer correctly but give no sign of explannation on which sentences the answer rests on. When they miss, models tend to answer anyway and invent plausible text. A good RAG model will instead reference the evidence it gave to its answer and withhold judgment if evidence is not available.

This is a natural reinforcement learning problem. The prompt, a question with k passages, is the first state; each of the token produced are an action; the episode ends at the `<answer>` tag, and a rule-based reward is given only then, scored against a reference answer and reference evidence. No reward model is trained, which keeps credit assignment transparent. Because the reward is terminal, the task can be considered as a bandit over whole responses, and critic-free methods as GRPO [4] can be used. We will use GRACE [1] which pairs a gated reward with a DAPO-style optimiser [3]. Begin the algorithm at a starting point and study the reward as well as the algorithm itself as variables which are subject to change.

2 Statement of the Problem

The setup of the problem is a sample s where s consists of a question q , its passage to gold C $_\,gt$, a distractor passage to confuse llm $C_\,dis$, the answer to gold a $_\,gt$ and the sentence for evidence for gold as $e_\,gt$. The retriever produces a context C_{ret} of k passages, and the policy generates $o \sim \pi_\theta(\cdot \mid p)$ for the prompt $p = P(q, C_{ret})$. The response can take one of the two paths: either an `<evidence>` path that quotes support when the ground truth is present in the retrieved contextippl $C_{gt} \subseteq C_{ret}$, or an `<llm>` path with a disclaimer when it is not. Training maximises $J(\theta) = \mathbb{E}[R(o, s)]$ for a reward R we design.

The policy can fail three ways: answer without sufficient evidence (hallucination), refuse when evidence was present (over-refusal), or pick the right path yet quote the wrong sentence. Any scalar reward fixes a trade-off between these, and the optimiser decides how reliably the policy reaches it. We ask:

RQ1 Does gating the reward rather than summing all terms, based on whether the query is answerable or not , stop the llm to guess on insufficient evidence?

RQ2 With the reward fixed, how do GRPO and DAPO compare in accuracy and sample efficiency?

3 Significance and Problem Statement

Retrieval-Augmented Generation (RAG) is a sequential decision-making problem. A policy must evaluate stochastic state observations like retrieved contexts. It then takes actions through text generation. In high-stakes fields like medicine or law, confident hallucinations are dangerous. An honest refusal is much safer. Therefore, teaching a model *abstention* is critical. The model must

recognize its epistemic boundaries and safely decline unanswerable questions.

This project is a strict testbed for *reward design* and *policy collapse* in Reinforcement Learning. The model faces a bifurcated action space. It must choose an `<evidence>` path to cite sources or an `<llm>` path to abstain safely. Poorly shaped scalar rewards inevitably induce reward hacking. The policy might collapse into always refusing or always guessing. We frame RAG as an episodic MDP to solve this. We will investigate how hierarchical gated rewards prevent this exploitation. We will also observe how on-policy algorithms balance exploration and exploitation across disjoint reasoning paths.

4 Approach

We will use two QA datasets a shown in (Table 1). From each question we will build cases that are answerable and unanswerable. Answerable samples will have the gold passages inserted if missing; unanswerable samples will have every gold passage replaced by a distractor. We will retrieve $k = 5$ passages using two retrievers, BM25 and the Qwen3-Embedding-0.6B dense model, and split each evenly.

Dataset	**Domain**	**Train**	**Test**	**Test ans : unans**
HotpotQA	Open-domain multi-hop (Wikipedia)	2,000	500	296 : 204
QASPER	Scientific-paper QA	2,593	1,451	1,200 : 251

Table 1: Two open source datasets. The answerable and unanswerable split is for the dense retriever at $k = 3$. HotpotQA will be our primary dataset; QASPER will serve as an optional cross-domain test.

We aim to use two reinforcement-learning algorithms that is GRPO and DAPO. GRPO (Group Relative Policy Optimization) [4] eliminates the need for a separate critic model by scoring a group of sampled responses and judging each rollout relative to the group average. DAPO [3] is an improvement of GRPO (decoupled clipping and dynamic sampling).

We will start from the reward function mentioned in the GRACE paper and then try our reward variation against it. The GRACE reward is as follows:

$$R = \begin{cases} 0, & \text{invalid format} \\ R_f, & \text{valid format, wrong path} \\ R_f + R_p, & \text{correct } \texttt{<llm>} \text{ path} \\ R_f + R_p + \alpha\,\mathrm{RL}_{ev} + \beta\,\mathrm{RL}_{ans}, & \text{correct } \texttt{<evidence>} \text{ path} \end{cases}$$

where R_f=0.5 rewards valid output tags, R_p=1 rewards choosing the right path (answer vs. abstain), and $\mathrm{RL}_{ev}, \mathrm{RL}_{ans}$ are the ROUGE-L overlap of the quoted evidence and the final answer with the gold references.

For the reward training, we will monitor the reward hacking by tracking the share of responses on each path, quoted-evidence length, and answers that copy the question. If necessary, deploy necessary gaurdrails to avoid reward hacking.

5 Framework

We will use the same commit of **verl** as used by GRACE, which will be pinned; GRACE trains with verl's **DAPO** recipe and his existing GRPO and PPO recipes will provide us with the two optimisers; we do not write new algorithm code to compare. **vLLM** will generate rollouts and serve models during data generation and evaluation. We will follow the environment used by the repository: Python 3.12, CUDA 12.4, and NVIDIA GPUs, with a LoRA path to keep training

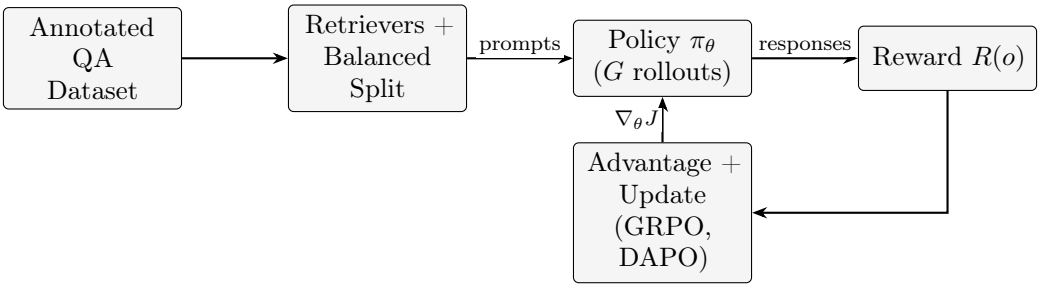


Figure 1: Training loop; the reward and the optimiser are swapped independently.

on the lighter weight model. We will write our three rewards as small Python functions and unit-test them on toy batches. We will be using **Rouge-L** and **BLEU** from the **evaluator.py** of the repository again for evaluation. The results of the models, along with the test set used, are displayed in the table below. **ARENA's** code and **DeepSeek API** prompt are used to perform using **LLM-as-a-judge**. We model the task as an episodic MDP (Table 2).

RL element	**Instantiation**
State	Prompt (instructions, question, k passages) plus tokens so far
Action	Next token from the vocabulary
Transition	Deterministic: append the token
Episode	Ends at end-of-sequence or a fixed token limit
Reward	Terminal, rule-based $R(o)$; zero at intermediate steps
Policy	π_θ, a Qwen3-1.7B model (LoRA)
Baseline	Group mean (GRPO) or learned value head (PPO)

Table 2: The task as an episodic MDP.

6 Metrics for Validation

We divide the test set into *answerable* (evidence retrieved) and *unanswerable* (evidence missing) subsets, defining these metrics:

- **Answerable Queries:** Exact Match (EM), BERTScore F1, and LLM-as-a-Judge (LJ) to evaluate semantic equivalence and evidence grounding.
- **Unanswerable Queries:** Abstention Accuracy (True Negative Rate), measuring queries correctly routed to the `<llm>` refusal path.
- **Aggregate (Primary Metric):** *Balanced Accuracy*, the arithmetic mean of both subsets, which penalises policies collapsing into a single path.

Baselines: Zero-shot models (e.g., Qwen3-4B) struggle with epistemic boundaries (∼42% abstention on QASPER, ∼17% EM on HotpotQA). Supervised Fine-Tuning improves formatting but remains brittle. Recent agentic models (e.g., Search-R1) plateau at 63%–67% overall accuracy due to high hallucination rates.

Proposed RL Improvement: We propose a cost-sensitive RL formulation with a multi-stage gated reward to explicitly penalise false-positive hallucinations. We aim to demonstrate improved sample efficiency and stability over standard DAPO, achieving quantifiable improvements over zero-shot, SFT, and agentic baselines.

State-of-the-Art (SOTA): The GRACE framework [1] sets the SOTA (74% accuracy on QASPER, 81% on HotpotQA). We use these figures as an optimal benchmark.

7 Team Members

Name	Work Assigned
Aadit Maniar	Data and retrieval pipeline: balanced split construction, API integration, adversarial distractor setup.
Dakshi Goel	Reward design: implementation and unit-testing of R0, R1, and R2, plus reward-hacking audits.
Chaitanya Bagul	RL training and evaluation: GRPO/DAPO execution, tracking metrics, logging curves, and error analysis.

Table 3: Team roles and work assignment.

References

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